

Beta-Two Dyson–Ornstein–Uhlenbeck Diffusion: Matrix Radialization and Ordered GUE Equilibrium

HCS-C378 theorem package

4 September 2026

Abstract

We fix a variance normalization for Hermitian Ornstein–Uhlenbeck diffusion and carry it without rescaling through the ordered eigenvalue process. Trace-metric matrix Brownian motion with drift $-H/2$ gives unit eigenvalue noise, unit reciprocal-gap repulsion, and ordered GUE equilibrium. Direct second-order perturbation yields the chamber generator, and its drift is one half the logarithmic gradient of the ordered GUE density. This first round closes matrix radialization, normalization, and reversibility only.

中文摘要

本文从厄米矩阵空间的迹度量出发，为奥恩斯坦乌伦贝格扩散固定唯一的方差约定，并将该约定逐项传递到递增排列的特征值过程。矩阵布朗运动的对角分量给出单位二次变差，非对角分量的二阶扰动则产生单位倒间隙排斥，线性回复漂移保持负二分之一的系数。由此得到有序腔室内的精确戴森型随机微分方程及其生成元，没有再用矩阵元尺度或时间重参数作隐性修正。生成元漂移恰为有序高斯酉系综密度对数梯度的一半，因此该密度可逆；其归一化常数由完整厄米积分和酉共轭分解确定。特别地，单维边界退化为普通一维回复过程，而任意有限维的所有系数都由同一迹度量产生；这说明可逆律不是按外部目标谱拟合得到的数值现象。本轮的全部结论止于矩阵径向化、系数锁定与平衡分布。

Keywords: Hermitian matrix diffusion; eigenvalue process; Ornstein–Uhlenbeck dynamics; ordered chamber; GUE equilibrium

关键词：厄米矩阵；特征值扩散；奥恩斯坦乌伦贝格过程；有序腔室；高斯酉系综

1 One normalization from matrix to chamber

Let Herm_N carry the real trace inner product $\langle A, B \rangle = \text{Tr}(AB)$. Standard Brownian motion $\mathcal{B}_t$ in this Euclidean space drives

$$dH_t = d\mathcal{B}_t - \frac{1}{2}H_t dt. \quad (1)$$

The trace-orthonormal basis consists of E_{ii} and, for $i < j$,

$$\frac{E_{ij} + E_{ji}}{\sqrt{2}}, \quad \frac{i(E_{ij} - E_{ji})}{\sqrt{2}}. \quad (2)$$

Thus every diagonal matrix martingale in an instantaneous eigenbasis has quadratic variation dt , while every off-diagonal complex entry has expected squared modulus dt . This sentence fixes the factor that is often hidden in matrix-entry conventions.

Write

$$W_N = \{x \in \mathbb{R}^N : x_1 < \cdots < x_N\}, \quad h(x) = \prod_{i < j} (x_j - x_i), \quad d = \frac{N(N-1)}{2}. \quad (3)$$

Theorem 1 (Radialization in the trace convention). *If H_0 has simple spectrum, then, up to their first collision, its increasingly ordered eigenvalues solve*

$$dX_i = dB_i + \sum_{j \neq i} \frac{dt}{X_i - X_j} - \frac{X_i}{2} dt, \quad (4)$$

where the B_i are independent standard real Brownian motions. The generator on W_N is

$$\mathcal{L} = \frac{1}{2} \sum_i \partial_i^2 + \sum_i \left(\sum_{j \neq i} \frac{1}{x_i - x_j} - \frac{x_i}{2} \right) \partial_i. \quad (5)$$

Proof. For a normalized eigenvector u_i , second-order Hermitian perturbation gives

$$dX_i = \langle u_i, dH u_i \rangle + \sum_{j \neq i} \frac{|\langle u_j, d\mathcal{B} u_i \rangle|^2}{X_i - X_j}. \quad (6)$$

The covariance statement following (2) turns the first martingale term into dB_i , the second-order term into the displayed reciprocal gaps, and the matrix drift into $-X_i dt/2$. This proves the formula until the first collision. $\square$

Theorem 2 (Ordered GUE reversible law). *The probability density*

$$\pi_N(x) = Z_N^{-1} e^{-|x|^2/2} h(x)^2 1_{W_N}(x), \quad Z_N = (2\pi)^{N/2} \prod_{j=0}^{N-1} j!, \quad (7)$$

is reversible for (5).

Proof. The drift is one half of the logarithmic gradient:

$$\frac{1}{2} \partial_i \log(e^{-|x|^2/2} h^2) = -\frac{x_i}{2} + \sum_{j \neq i} \frac{1}{x_i - x_j}. \quad (8)$$

Hence $\mathcal{L}f = (2\pi_N)^{-1} \nabla \cdot (\pi_N \nabla f)$, where π_N denotes its density, proving symmetry. For the normalizer, use $h = \det[\text{He}_{i-1}(x_j)]$ and scalar Hermite orthogonality in Andréief's identity. The integral over $\mathbb{R}^N$ is $N!(2\pi)^{N/2} \prod_{j=0}^{N-1} j!$. The integrand is symmetric, so the ordered chamber contributes $1/N!$ of that value. The invariant matrix density is proportional to $e^{-\text{Tr } H^2/2}$; Weyl integration gives the same ordered density. $\square$

The $N = 1$ face is ordinary scalar Ornstein–Uhlenbeck diffusion. No large- N passage or entrance from a multiple eigenvalue is included.

References

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round zero matrix radialization and reversible GUE density